

Week 3 Internship Documentation

Malware Detection and Security Monitoring using pfSense, Squid Proxy, ClamAV, and Wazuh SIEM

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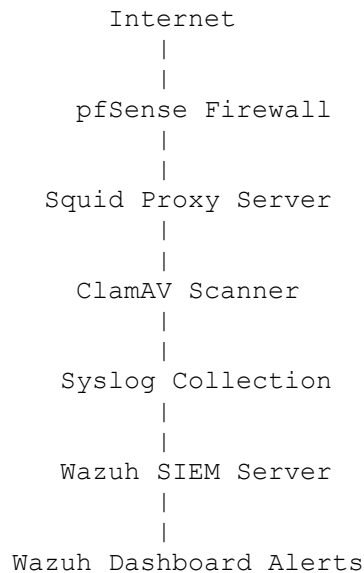
1. Project Objective

The objective of this project was to deploy and configure a security monitoring environment capable of detecting malicious activities using firewall monitoring, proxy inspection, antivirus scanning, and SIEM-based alerting.

The implementation included:

- pfSense Firewall
- Squid Proxy
- ClamAV Antivirus Engine
- Syslog Integration
- Wazuh SIEM
- Custom Wazuh Decoders and Rules
- Malware detection testing using EICAR

2. Lab Architecture



3. pfSense Firewall Deployment

Configuration

pfSense was deployed as the network firewall and configured to:

- Manage network traffic
- Monitor firewall events
- Forward logs to the SIEM server

Remote Logging Configuration

pfSense was configured to send security logs to the Wazuh server through syslog.

Configuration:

```
Status
→ System Logs
→ Settings
→ Remote Logging Options
```

Configured:

- Remote Syslog Server: Wazuh Server IP
 - Protocol: UDP
 - Port: 514
-

4. Squid Proxy Configuration

Squid Proxy was deployed to:

- Monitor HTTP/HTTPS traffic
- Control web access
- Generate proxy access logs

Functions implemented:

- Proxy request monitoring
- URL logging
- Traffic inspection

Logs generated:

```
/var/log/squid/access.log
```

5. ClamAV Integration

ClamAV was configured as the malware scanning engine.

Purpose:

- Detect malicious files
- Scan downloaded content
- Generate malware detection events

ClamAV components:

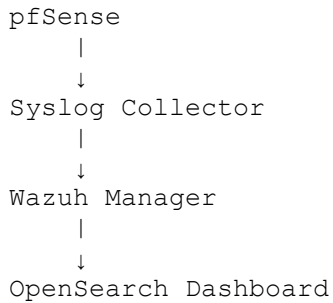
```
clamd  
freshclam
```

Virus database updates were enabled to maintain current detection capability.

6. Syslog Integration

Security logs were forwarded to Wazuh through syslog.

Log flow:



7. Wazuh SIEM Configuration

Wazuh was deployed for:

- Log collection
- Threat detection
- Security alert generation
- Event correlation

8. Custom Wazuh Decoder

A custom decoder was created for pfSense DHCPv6 events.

Decoder:

`pfsense-dhcp6c`

Purpose:

- Identify pfSense DHCPv6 logs
- Extract important fields
- Normalize events for detection

9. Custom Wazuh Rule

A custom detection rule was created:

Rule ID: 100100

Description:
pfSense DHCPv6 Client Activity

Level:
5

The rule successfully generated alerts in Wazuh.

Example detected event:

pfSense DHCPv6 Client Activity
Sending Solicit

10. EICAR Malware Detection Test

The EICAR test file was used to validate malware detection capability.

EICAR is a harmless industry-standard antivirus testing file.

Testing process:

1. EICAR test file was downloaded through the proxy.
2. Squid logged the request.
3. ClamAV scanned the file.
4. Malware signature was detected.
5. Security logs were forwarded to Wazuh.
6. Wazuh generated an alert.

Detection flow:

```
EICAR File
  |
  ↓
Squid Proxy
  |
  ↓
ClamAV Detection
  |
  ↓
Syslog
  |
  ↓
Wazuh Alert
```

11. Alert Verification

Alerts were verified using:

- Wazuh Dashboard
- OpenSearch queries
- Log analysis

Example:

```
Rule ID: 100100
Decoder: pfsense-dhcp6c
Source: pfSense
```

Incident Response Plan

Malware Detection Incident Response Plan

1. Identification

Objective:

Detect and confirm malicious activity.

Detection Sources:

- Wazuh alerts
- ClamAV malware detection
- Squid proxy logs
- pfSense firewall logs

Analyst Actions:

1. Review Wazuh alert details.
2. Identify:
 - Source IP
 - Destination URL
 - File information
 - Timestamp
 - User/device involved
3. Verify whether the event is a true positive.

2. Containment

Objective:

Prevent further spread.

Actions:

- Block malicious IP/domain through pfSense.
- Isolate affected endpoint.
- Disable suspicious network activity.
- Block malicious URLs through Squid Proxy.

3. Investigation

Objective:

Understand the attack.

Analyst reviews:

Firewall Logs

Check:

- Source address
- Destination address
- Ports
- Connection attempts

Proxy Logs

Analyze:

- Requested URLs
- Download activity
- User activity

ClamAV Logs

Review:

- Detection name
- File information
- Scan result

Wazuh Logs

Analyze:

- Timeline
 - Related alerts
 - Indicators of Compromise (IOCs)
-

4. Eradication

Actions:

- Remove malicious files.
 - Run complete antivirus scans.
 - Update malware signatures.
 - Patch vulnerable systems.
 - Remove persistence mechanisms.
-

5. Recovery

Actions:

- Restore normal network access.
 - Monitor affected systems.
 - Verify no additional malicious activity exists.
 - Continue SIEM monitoring.
-

6. Lessons Learned

After the incident:

- Update firewall rules.
- Improve detection rules.
- Update antivirus signatures.
- Review user security awareness.
- Improve monitoring coverage.

Conclusion

This project successfully implemented a SOC monitoring environment using pfSense, Squid Proxy, ClamAV, and Wazuh SIEM. The system was capable of collecting security logs, detecting malware activity through EICAR testing, generating SIEM alerts, and supporting incident response activities.